What do you want to learn?

ALICE L WALTON SCHOOL OF MEDICINE

Pharmacodynamics1: Receptor Agonists and Antagonists

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Adapted from: Receptor Agonists and Antagonists

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Learning Objectives (5) Versions

After completing this brick, you will be able to:

1. Describe the downstream signaling pathways activated upon receptor engagement, using examples of clinically relevant drug-receptor interactions.
2. Interpret quantitative dose-response data to deduce drug attributes such as pote ical implications.
3. Classify agonists into full, partial, and inverse agonists, differentiating them based on their potency, efficacy, and intrinsic properties.
4. Classify antagonists into competitive and noncompetitive, explain the clinical significance and predict their effects on agonist dose response curves.
5. Analyze Lineweaver-Burk or Michaelis-Menten plots, extracting vital enzyme kinetic parameters and determining the specific inhibition type – competitive or non-competitive.

CASE CONNECTION

OP is a 50-year-old man who is going to undergo minor surgery for his knees. He is diagnosed with hypertension and has a history of a heart attack (5 years ago). OP is currently prescribed Lisinopril (hypertension-medication), Metoprolol (Arrhythmia-control his elevated heart rate), and Aspirin (reducing clotting risk). You are a surgical resident meeting with OP 3 weeks prior to his surgery to review OP's medication history. You state that OP is supposed to stop Aspirin 1 week before the surgery and restart it post-discharge. You state that Aspirin can increase the risk of bleeding. OP is very confused. He conveys that he is concerned about stopping the Aspirin 7 days before and not a day before the surgery.

What will you explain to OP as to why he has to stop his Aspirin 7 days prior to the surgery?

Consider your answe of the brick.

WHAT IS PHARMACODYNAMICS?

An easy way to define pharmacodynamics is what the drug does to the body. For example, a drug might bind to a target (such as a receptor) to set off a cascade of cellular changes, or it might alter the chemical conditions inside the body (e.g., by adding a base such as bicarbonate). In addition, drugs can either activate or inhibit certain physiological effects. Pharmacodynamics is the relationship of drug concentration at the site of action, including therapeutic and adverse effects. How much of a drug is required to achieve the desired effect is another aspect of pharmacodynamics. In the next sections, we are going to discuss where the drugs bind.

What Are Receptor Agonists and Antagonists? (LO#1)

In order for the drug to elicit its effects, it has to bind or attach to a component of the organism. In most cases, pharmacologists refer to this component as a receptor. The receptor is a component of a cell or organism that interacts with a drug and initiates the chain of events leading to the drug's observed effects. Receptors are defined as proteins that recognize a signaling molecule that is endogenous to the cell or organism which it then translates into a functional cellular event. These receptors determine the relationship between dosing a drug and its effect, determine the selectivity of how the drug works, and mediate the action of the drug. Receptors are cell membrane bound or recognize signals from the cell surface. Receptor types include G-protein-coupled receptors (G yme-linked (i.e., catalytic) receptors (Tyrosine Kinase receptors), other membrane-associated receptors, and nuclear receptors.

The same principles can be applied to how a drug activates or inactivates a receptor, one of the main mechanisms of drug action in the domain of pharmacodynamics (how drugs act on the body). Ligand is a term that is used to describe a molecule that binds to a receptor. Ligands can be endogenous or exogenous to the body. An agonist binds a receptor and "activates" (or induces) a cellular action. Antagonists blocks a receptor function and its associated cellular action by binding or competing for binding (Figure 1).

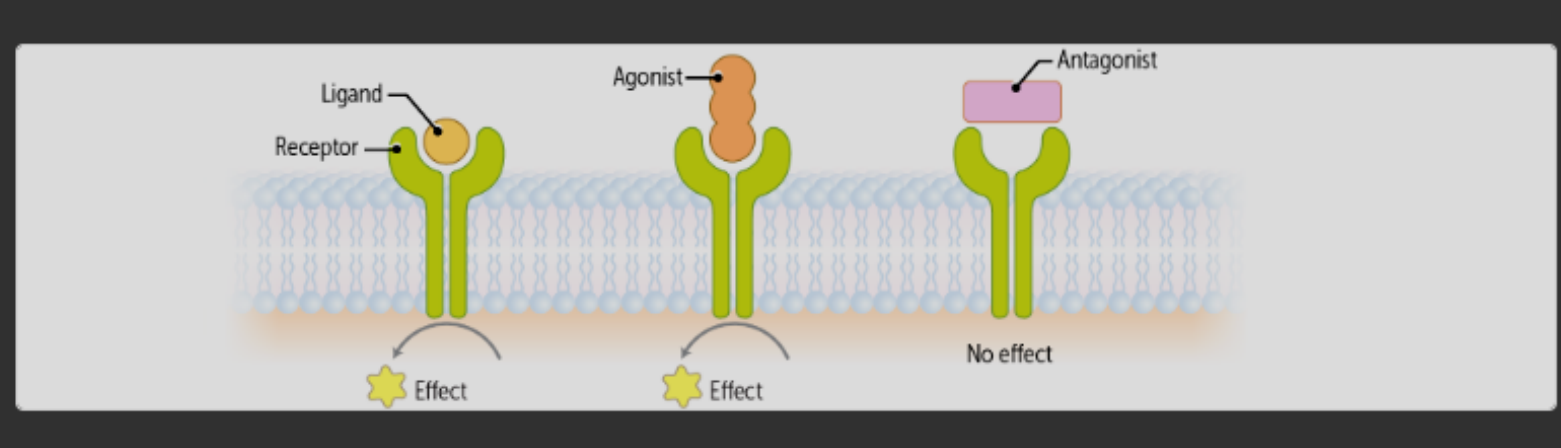

Figure 1

CLINICAL CORRELATION

The β_2-receptor agonist albuterol is used in the treatment of asthma because it activates β_2 receptors in the lungs to cause bronchodilation, opening the airways.

What is the role of an agonist in pharmacodynamics?

What is a Dose-response relationship? (LO#3)

First, lets look at a dose-response relationship curve. You are going to see these plots often as we discuss pharmacodynamics.

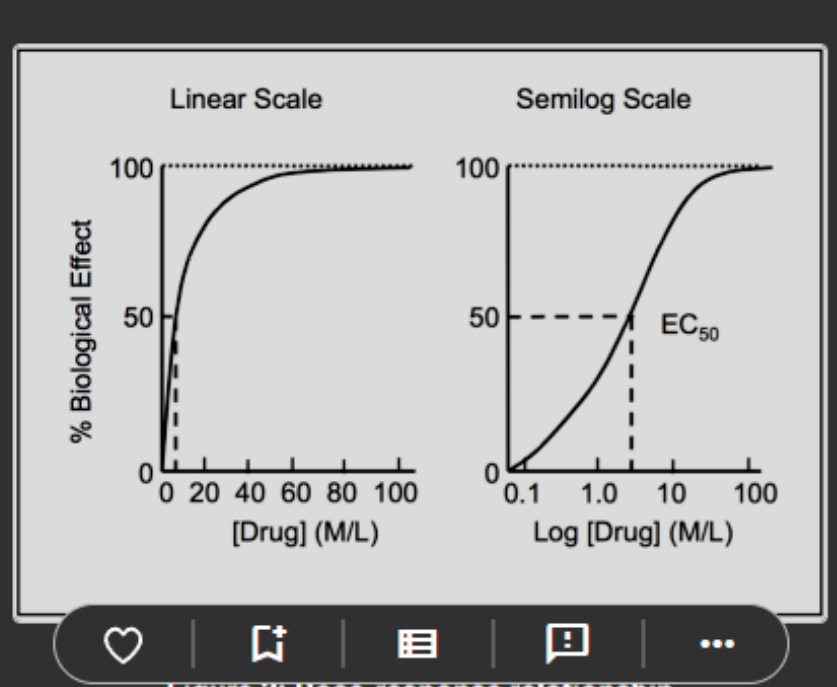

Figure 2: Dose-response relationship

When plotted on a linear scale (left panel), a dose-response relationship (Figure 2) is hyperbolic. As discussed in the enzyme kinetics brick in BOOM, at high concentrations the response reaches a maximum due to saturation of available receptors by the drug. However, when plotted on a semi-log scale (logarithm of drug concentration vs. effect), the relationship becomes sigmoidal (S-shaped). The semi-log plot is the preferred method for plotting dose-response relationships because it becomes easier to accurately determine the EC_{50} value (the concentration which produces 50% of the maximum response). We will discuss this concept later, but you should note that it is very important to determine a drug's EC50 by placing it on a linear portion of the curve.

WHAT ATTRIBUTES OF A DRUG CAN DEFINE IT'S BINDING TO A RECEPTOR? (LO#3)

Drug effects

Knowing how a drug interacts with a target is important, but how do we know if a drug is working? This is where the drug's effect comes in. A drug that is supposed to neutralize stomach acid should have the effect of increasing stomach pH, for instance. If a drug is supposed to activate dopamine receptors, its effect should be to increase G protein activity. If, on the other hand, the drug is supposed to inhibit dopamine receptors, it should decrease G protein activity. Basically, drug effect describes the downstream consequences that occur after a drug-tar

Drug efficacy

Figure 3 shows increasing drug concentrations for two drugs. The x axis is the drug concentration, plotted in logarithmic form, and the "effect" is measured on the y axis. This is called a **graded dose-response curve.** The drug effect increases in proportion to the increase in drug dose. Parameters associated with a graded dose-response curve are described below.

Efficacy and Potency

Efficacy is a measure of the maximal response produced by the drug. The maximum effect (red lines) of the drug is called the **Emax**. The drug represented by the blue line has a higher Emax value than the drug represented by the green line; therefore, it has more efficacy.

We can also compare drugs to determine how much of each will produce a certain effect. To do this, we look at the concentration at which a drug produces its half-maximal effect, EC50, or 50% of Emax (yellow lines in Figure 9). The drug represented by the green line is thus more potent than the drug represented by the blue line because it achieves its EC50 at a lower concentration (less drug is required for it to act).

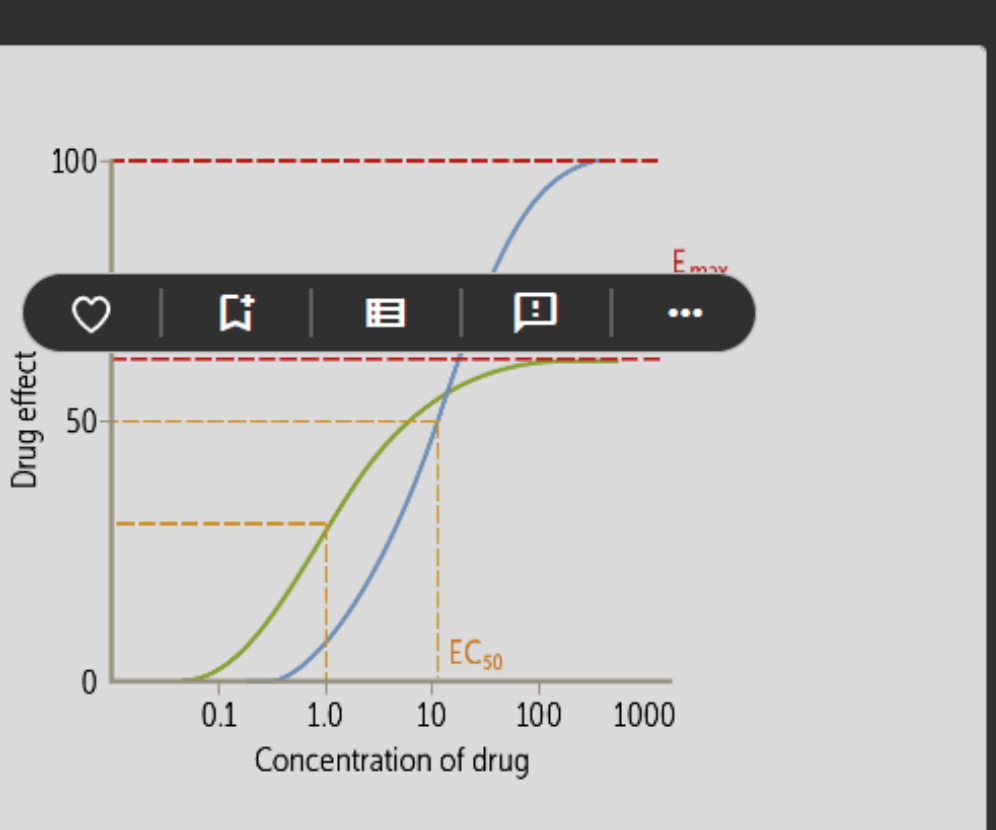

Figure 1: Emax and EC50 of a drug

The EC_{50} of a drug is used to determine its **potency.** Potency is a measure of how much drug is needed to reach an established effect, and a standard measurement is the EC_{50}.

Selectivity

Two other terms are important to note. If a drug has one effect, and only one effect on all biological systems, it possesses the property of **specificity**. But in reality, the vast majority of drugs are **selective** rather than specific. The majority of drugs are selective because most drugs will act on more than one receptor site once they reach an appropriately high concentration. In clinical practice, you will often hear the term selective agonist (or antagonist) or non-selective agonist (or antagonist). For example, both Propranolol and Metoprolol are β-blockers. They are used to inhibit increases in heart rate that underline many cardiovascular diseases. However, Propranolol, is known as a non-selective β-blocker (blocks all β receptors), whereas Metoprolol is a *selective* β1-blocker. e, it is often used to in cases of increased heart rate (or tachycardia).

Affinity

Affinity can be defined as the extent or fraction to which a drug binds to receptors at any given drug concentration or how 'tightly' the drug binds to the receptor. Affinity is one of the factors that determine potency. (Figure 4)

Affinity is inversely proportional to the potency of a drug (1/Kd), where Kd is the dissociation constant. The strength of the binding (interaction) of a ligand (or drug) and its receptor can be described by affinity. The higher the Kd value, the weaker the binding and the lower the affinity. With a smaller Kd value, the greater the affinity of the ligand or drug for its target.

For example, albuterol, used in rescue inhaler for asthma, has a Kd of 100 nanomolar (nM) for the beta (β)-2 receptor.

Higher affinity often increases potency. Higher affinity→ more receptors occupied (bound) at lower concentration → EC50 decreases → potency increases (Note: This relationship is not always applicable to all drug)

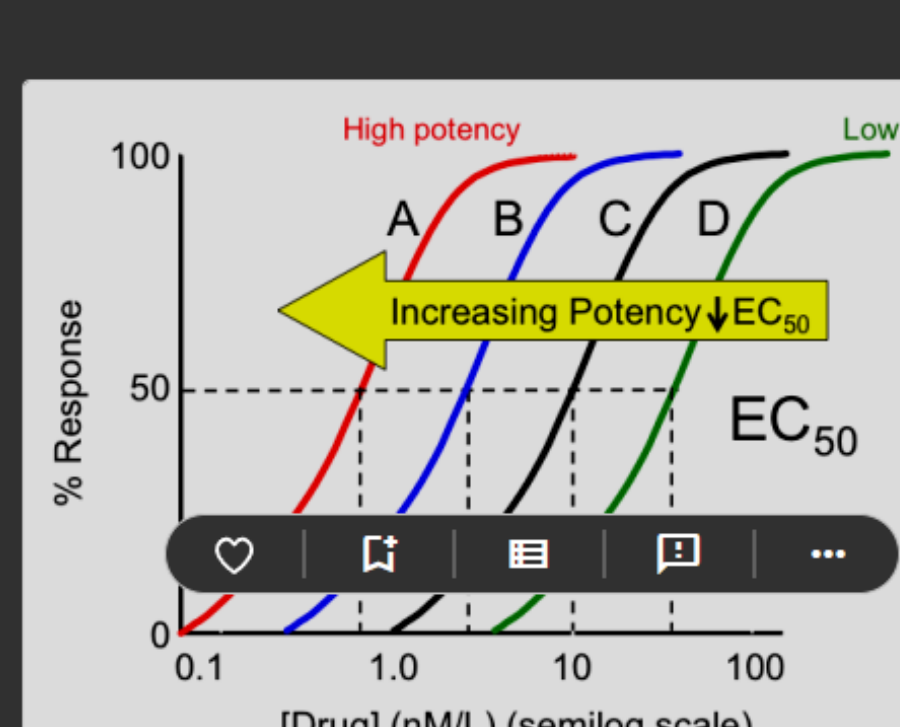

Figure 4: Relationship between potency and affinity

Drugs can bind to the same receptor but might differ in terms of their affinity and/or efficacy. Drugs that can produce an effect at a lower concentration are " more potent". In Figure 4, drug D is the least potent since it take more concentration to produce its EC50, whereas drug A can produce the same response at lower concentrations.

The following question must be completed before continuing

ACTIVE LEARNING QUESTION

Match the definitions of drug-binding to the property

Emax Efficacy Affinity Potency

Maximum effect of the drug

Maximal response of a drug

Extent of drug binding

Amount needed for EC50

response at lower concentrations.

The following question must be completed before continuing

ACTIVE LEARNING QUESTION

Match the definitions of drug-binding to the property

Emax Efficacy Affinity Potency

Maximum effect of the drug

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Amount needed for EC50

Explanation (requires correct answer)

Continue to Next Section